

Narrative Chain-of-Thought: Inference-Time Scaffolding for Defeasible Ethical Reasoning in Large Language Models

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Abstract

Standard chain-of-thought on moral dilemmas exhibits two structural failure modes that matter for deployment: *stakeholder collapse* (the multi-party situation is reduced to a binary) and *uncertainty suppression* (the model commits to a projected outcome it cannot warrant). We introduce *narrative chain-of-thought* (N-CoT), a system prompt that asks the model to name a protagonist, enumerate stakeholders, project two-step consequences, articulate genuine uncertainty, and only then commit. The protocol adds no training data, parameters, or fine-tuning. On a 100-scenario sample of DailyDilemmas across four generators from three vendors, N-CoT cuts stakeholder collapse from up to 31% to under 1% and uncertainty suppression from up to 72% to 1–24%; under length residualisation the effect remains large on the OpenAI and xAI generators but is length-mediated on both Anthropic generators (Cliff’s δ confidence intervals span zero). A section-by-section ablation gives the prompt-side contribution a clean mechanistic shape: each substantive section is the specific carrier of its target metric, with negligible cross-variable spillover. A panel of seven length-invariant compression and entropy proxies fails to validate the underlying parsimony hypothesis at inference time, an honest negative result we report rather than read past. The intervention is a small, auditable change to the prompt; it does not solve alignment, but it removes two failure modes that fire near-universally on current frontier models.

1 Introduction

A frontier LLM asked to think step by step about an everyday moral dilemma frequently produces a trace that does not look like ethical reasoning at all. When asked “think step by step” on the DailyDilemmas corpus (Chiu et al., 2025), four

frontier generators spanning three vendors (OpenAI, Anthropic, xAI) and two model tiers suppress articulated uncertainty on 50–72% of scenarios and collapse the stakeholder set on 15–31% of scenarios. This is a structural deficit of the trace, not a labelled safety violation, and the deployment-relevant failures that have been reported elsewhere (GPT-4o sycophancy rollback (OpenAI, 2025b,a), agentic-misalignment blackmail rates up to 96% in simulated deployments (Lynch et al., 2025), reasoning-induced misalignment (Yan et al., 2025)) are coherent with the same underlying picture of optimising locally rewarded behaviour without articulating downstream consequences (Sharma et al., 2024; Malmqvist, 2024). We do not claim N-CoT reduces these deployment-relevant failures on the current deployment-relevant benchmarks: SycophancyEval is saturated on the quartet and the two replicated agentic scenarios refuse unconditionally on every model under either scaffold (full results §7). The contribution this paper makes is on the upstream trace, not on the saturated downstream probes; the hypothesis that fixing the upstream trace propagates to harder, less-public deployment scenarios is registered as future work.

The hypothesis we test on the upstream trace is that asking the model to narrate the same dilemma, by naming a protagonist, enumerating stakeholders, projecting two-step consequences, and surfacing genuine uncertainty before committing, elicits a causally richer trace because it puts the model on the manifold of human narrative its pretraining corpus most heavily samples. Figure 1 states the mechanism intuition: the same dilemma flows through two different scaffolds, and the one that requires the model to narrate before committing eliminates both failure modes prompt-side. The methodological move is the one Gottweis et al. (2025) make for scientific discovery and Lu et al. (2024) extend to end-to-end research workflows:

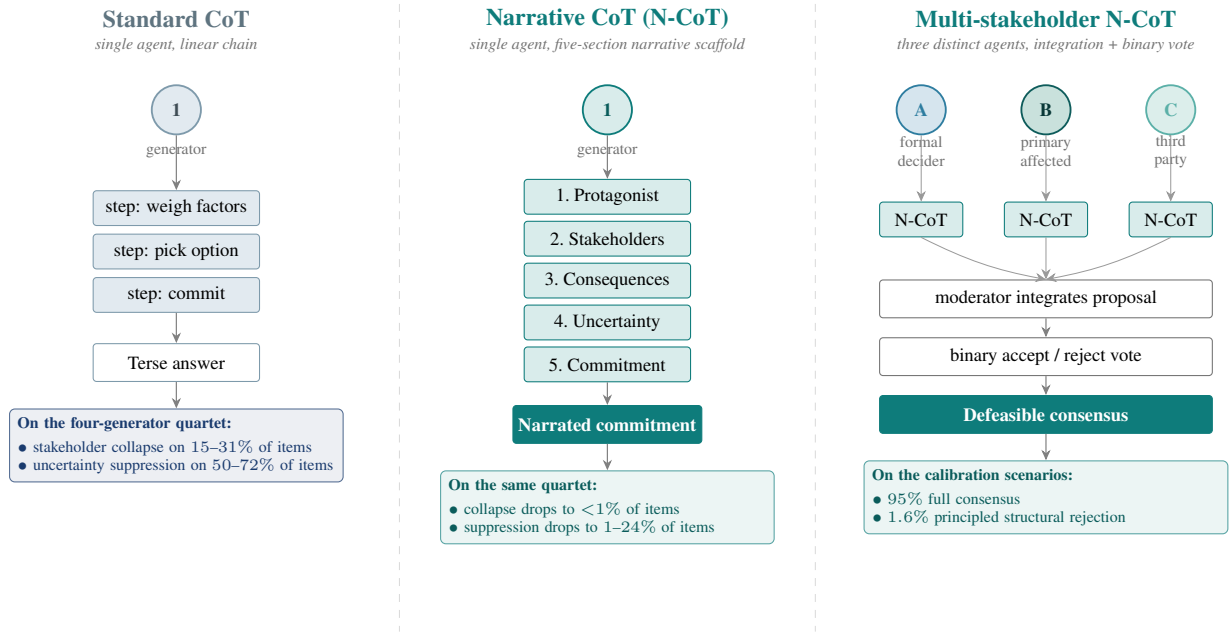


Figure 1: One scaffold, three settings. *Standard CoT* (left) lets one generator run a linear reasoning chain over the dilemma; the two failure modes diagnosed in §4 fire on this scaffold. *Narrative CoT* (middle) keeps the single generator but forces the trace through five narrative primitives (protagonist, stakeholders, consequences, uncertainty, commitment) before any commit; on the four-generator quartet this drops stakeholder collapse to <1% and uncertainty suppression to 1–24% of items. *Multi-stakeholder N-CoT* (right) reuses N-CoT as the per-agent generator and adds three distinct stakeholder narrators (formal decider, primary affected party, third party) whose modification requests are integrated by a moderator and then put to a binary accept/reject vote; on the calibration scenarios this protocol yields 95% full consensus with 1.6% principled structural rejection (§5). The same scaffold composes from one narrator to many.

scaffold LLM reasoning into the primitives of the target domain rather than treat the model as a monolithic question-answerer. The primitives we reify are the primitives of human ethical reflection (named agents, stakeholders, projected consequences, uncertainty, commitment), and at the multi-agent layer we replace the AI co-scientist’s meta-review role with a moderator that integrates competing stakeholder positions into a single proposal and then asks each stakeholder to vote on it (§5). The analogy is methodological and partial: hypothesis tournaments in scientific discovery can be ranked against a ground-truth signal (experimental falsification), whereas multi-stakeholder value standoffs do not have an equivalent ranking oracle, which is precisely why we report the integrated-proposal acceptance pattern and the residual structural rejections rather than a single “winning” value position.

Contributions. We diagnose two cross-vendor failure modes that empirically fire on standard CoT and replicate across the quartet with no single-vendor escape (§4). We show that N-

CoT eliminates both modes on the OpenAI and xAI generators under length residualisation, and that the effect on the two Anthropic generators is length-mediated (the length-residualised Cliff’s δ confidence interval spans zero on stakeholder count and on uncertainty score); a sub-instruction ablation on the flagship attributes each headline metric to the specific N-CoT section that carries it (§4.1). We stress-test the underlying parsimony intuition with a panel of seven length-invariant proxies for Algorithmic Causal Complexity, and report that every one of them collapses under length control (§6). We read this as falsification of the inference-time gzip claim rather than as confirmation, and frame the K_C apparatus as theoretical scaffolding for the training-time Interpreter Network (§9) where the supervision- signal claim can actually be tested. Two deployment-relevant floor effects are reported honestly: SycophancyEval is saturated on frontier models, and two agentic-misalignment replicates yield 0% harmful action across the quartet under either scaffold (§7), consistent with the pre-registered contingency that scaffolding cannot be measured against a failure

mode that does not fire. Two next steps are pre-registered: a Tier-3 human pairwise preference study (IRB pending) and the training-time Interpreter Network referenced above.

A pragmatic note. We are not claiming that N-CoT solves alignment, nor that it removes the need for training-time work. We are claiming that a minimal prompt-level intervention with a clear theoretical motivation produces large, length-controlled, cross-family shifts on the structural failure modes most directly tied to sycophancy and stakeholder erasure, and that the same scaffolding extended to multi-agent deliberation produces *defeasible* commitments, positions that revise under others’ narrated reasoning while holding firm where revision would materially harm a stakeholder.

2 Background and Related Work

Chain-of-thought and its failure modes. Chain-of-thought prompting (Wei et al., 2022; Kojima et al., 2022) elicits step-by-step intermediate reasoning before a final answer; it improves arithmetic and symbolic tasks but does not by construction force the trace to track situational structure. Recent analysis identifies a sharp tension: as chain-of-thought reasoning capacity grows, models become more responsive to harmful requests because reasoning and safety entangle in shared neuron populations, a phenomenon Yan et al. (2025) describe as “reasoning-induced misalignment”. N-CoT can be read as a constrained chain-of-thought that re-anchors the trace to a narrative-causal structure before the reasoning capacity is exercised; §4 shows that the intervention works on a current reasoning model as well as a non-reasoning one.

Narrative scaffolding for reasoning. Visualisation-of-Thought (Wu et al., 2024) shows that asking a model to externalise reasoning in the native representational format of a domain (spatial diagrams for spatial tasks) improves reliability. We apply the same principle to ethical reasoning: the native format is a story with named agents, projected consequences, and uncertainty (MacIntyre, 1981; Bruner, 1986). Empirical narrative-perspective interventions in medical decision contexts measurably increase perspective-taking and empathic concern (Shaffer et al., 2019; Bientzle et al., 2021, 2024), and recent work on generative agents shows long-horizon coherent behaviour is achievable from

narrated character state (Park et al., 2023).

Causal inference from narrative. The Computational Theory of Mind (Fodor, 1975; Putnam, 1967) treats understanding a sequence of events as inferring a structural causal model that explains them; the Kuleshov experiment (Kuleshov, 1974; Bordwell, 1985) is a textbook example. LLMs perform near random baseline on causal inference stripped of empirical pattern matches (Jin et al., 2024): pretraining supplies the ability to pattern-match narrative token sequences, not the structured causal reasoning that determines how those narratives resolve. The role of N-CoT is to convert that pattern matching into a usable causal trace at inference time.

Multi-agent debate and defeasibility. Irving et al. (2018) proposed AI safety via debate; Du et al. (2023) showed multi-agent debate improves factual reasoning. Both target task-correctness, not stakeholder-divergent value standoffs. Defeasible reasoning, the property that conclusions can be revised in light of new information without abandoning the underlying inference scheme, has a long lineage in moral philosophy (Pollock, 1987; MacIntyre, 1981). The protocol we evaluate in §5 stacks defeasibility on top of N-CoT and asks whether present-day models can be elicited to revise narrated commitments under narrated counter-evidence.

Multi-agent LLM systems for scientific discovery. A growing line of work scaffolds LLM reasoning into the primitives of a target domain rather than treating the model as a monolithic question-answerer. The AI co-scientist of Gottweis et al. (2025) casts hypothesis generation, peer review, ranking, evolution, and meta-review (the primitive moves of the scientific method) as specialised agents whose interactions, including a Generate-Debate-Evolve loop and an Elo-style ranking tournament, drive iterative quality improvement, yielding experimentally validated drug-repurposing and target-discovery proposals. The AI Scientist (Lu et al., 2024) closes the loop end-to-end (ideation, coding, experimentation, manuscript drafting, automated peer review). Domain-specific instantiations include Coscientist (Boiko et al., 2023) and ChemCrow (M. Bran et al., 2024) for chemistry, Agent Laboratory (Schmidgall et al., 2025) for ML research workflows, and the Virtual Lab (Swanson et al., 2025)

for interdisciplinary biomedical design with experimental validation. The common move across this family is to identify the activity that human experts already perform in the domain (hypothesise, critique, rank, integrate) and reify each activity as a co-operating LLM agent, with the meta-review or principal-investigator role synthesising across agents. Our two protocols ask the analogous question one level down: what are the primitives of human ethical reflection, and can a scaffold that reifies them as a structured trace (protagonist, stakeholders, projected consequences, uncertainty, commitment) and as multi-narrator deliberation (statement, rebuttal, integrated proposal, binary vote) deliver the same kind of inference-time quality lift on moral reasoning that the scientific-method scaffolds deliver on hypothesis generation? The multi-stakeholder protocol in §5 is particularly direct in this analogy: the moderator’s Round-4 integrated-proposal step is structurally the meta-review of [Gottweis et al. \(2025\)](#), applied to value-laden stakeholder positions instead of competing scientific hypotheses.

3 Narrative Chain-of-Thought

Definition. *Narrative chain-of-thought* (N-CoT) is a system prompt that instructs the model to produce, in first person, a five-section trace before any final answer: (1) characterise the protagonist (name, role, what they know); (2) enumerate the stakeholders whose lives intersect the decision and what is at stake for each; (3) project the consequences of each available action at least two steps out for each stakeholder; (4) state what remains genuinely uncertain about each projected future; (5) commit to a decision and explain why, within the narrative frame, that trajectory is preferable to the alternatives.

The intervention is a single change to the system prompt and does not add training data, parameters, or fine-tuning. The mechanism it is designed to exploit is that text matching the five-section structure is densely represented in the narrative subdistribution of the pretraining corpus (novels, scripts, case reports, ethnographic vignettes), so the model can amortise causal-trajectory inference by retrieval rather than by reasoning from first principles. Crucially, the prompt does not mention causal graphs, stakeholders’ utilities, or any formal apparatus. Algorithm 1 states the procedure precisely.

Algorithm 1 Narrative chain-of-thought (N-CoT): generation and structural coding for a single dilemma.

Require: dilemma s ; generator G ; judge J ; decoder Θ

Ensure: trace t and structural codes c

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1: Build N-CoT system prompt  $\pi_{\text{NCOT}}$  requesting five first-
   person sections:
2:   (1) PROTAGONIST: name role and epistemic state
3:   (2) STAKEHOLDERS: parties intersecting the decision
4:   (3) CONSEQUENCES: each action  $\geq 2$  steps forward
5:   (4) UNCERTAINTY: what remains genuinely unknown
6:   (5) COMMITMENT: chosen action and narrative war-
   rant
7: Generate trace
8:    $t \leftarrow G(\pi_{\text{NCOT}} \parallel s; \Theta)$ 
9: Code structural variables with judge  $J$ ’s rubric
10:   $(sc, mh, us, nf, ref, tr) \leftarrow J(t)$ 
11:  // stakeholder count, max causal hops, uncertainty
   score,
12:  // framework count, refused flag, truncated flag
13: Derive failure-mode tags (deterministic)
14:  STAKEHOLDERCOLLAPSE  $\leftarrow sc \leq 1$ 
15:  UNCERTAINTYSUPPRESSION  $\leftarrow us = 0$ 
16:  CONSEQUENTIALFLATTENING  $\leftarrow mh \leq 1$ 
17:  FRAMEWORKENUMERATION  $\leftarrow nf \geq 2$ 
18:  PREMATUREREFUSAL  $\leftarrow ref$ 
19: return  $(t, c)$ 

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3.1 Multi-Stakeholder Narrative Deliberation

A single N-CoT trace commits one protagonist to a position. Many real deployment-relevant decisions involve multiple narrators with conflicting stakes. We extend N-CoT into a four-round debate protocol that re-uses N-CoT as the per-agent generator. For each scenario, three stakeholder perspectives (formal decider, primary affected party, third party) each produce an N-CoT statement (Round 0). They then exchange (Round 1, rebuttal), restate (Round 2, final position), respond to a moderator-constructed synthesis with one of ACCEPT, ACCEPT_WITH_MODIFICATION, or REJECT (Round 3), and finally cast a binary ACCEPT/REJECT vote on a single integrated proposal that the moderator constructs from the three Round-3 modification requests (Round 4). The moderator is a separate LLM, cross-family with at least one generator. The protocol’s core hypothesis is that the integration step in Round 4 makes *defeasibility* measurable: an agent who continues to reject a proposal that addresses its modifications is signalling a structurally non-negotiable concern, not procedural friction.

The architectural debt is to the multi-agent systems for scientific discovery surveyed in §2. The AI co-scientist of [Gottweis et al. \(2025\)](#) reifies the activities of hypothesis generation, peer review,

ranking, evolution, and meta-review, the primitive moves of the scientific method, as distinct agents whose interactions raise the quality of the joint output. Our protocol takes the same compositional move and applies it to ethical reflection rather than scientific discovery: stakeholder narrators replace hypothesis generators, narrated rebuttal replaces the reflection agent, and the moderator’s integrated-proposal-plus-binary-vote step (Round 4) plays the role of meta-review, but against value-laden positions rather than competing scientific hypotheses. The motivation for treating ethical reflection as a multi-agent activity is that values that look incompatible from a single protagonist’s vantage often have an integration the protagonists themselves can endorse once each side’s non-negotiable stake is on the table, the same way the AI co-scientist’s ranked-and-evolved hypotheses converge on candidates no single generator agent would have produced.

4 Experiment 1: N-CoT at Scale

Setup. Three prompting conditions are compared: bare input/output, standard CoT (“Think step by step, then give your answer”), and N-CoT (the five-section structure of §3, with no formal apparatus mentioned). Decoding parameters are held identical across conditions. Four generators are tested: gpt-5.4-nano ($N=20$) from the original pilot, plus three new generators spanning two additional vendors: claude-haiku-4-5 ($N=20$), grok-4-1-fast-reasoning ($N=5$), and the flagship claude-sonnet-4-6 ($N=5$, cost-controlled). Here N is the number of independent samples per (scenario, condition, generator) cell drawn at temperature 1; with the 100-scenario sample and three conditions this yields $300N$ generations per generator before the cross-judge coding stage. Coded sample sizes (N_S, N_N in Table 1) are slightly smaller because the rubric drops generations where the judge could not extract a coded variable. Outputs are coded by two cross-vendor judges (claude-haiku-4-5 primary, gpt-5.4-nano secondary) on a six-variable rubric with quadratic-weighted Cohen’s κ (Cohen, 1960) per variable. The scenario pool is a 100-scenario stratified sample of DailyDilemmas (Chiu et al., 2025), spanning 18 topic groups; the same deterministic stratified sample (seed 42) is used for all four generators, making the scenario sets comparable.

Generator (vendor, tier)	N_S	N_N	Δ_{SC}	Δ_{US}
gpt-5.4-nano (OpenAI, b)	1,989	1,985	−30	−72
claude-haiku-4-5 (Anth., b)	1,999	1,996	−25	−29
grok-4-1-fast-r. (xAI, b)	515	515	−13	−44
claude-sonnet-4-6 (Anth., f)	500	500	−26	−37

Table 1: Cell sizes and percentage-point drops for the two failure modes that empirically fire (Δ_{SC} = stakeholder collapse, Δ_{US} = uncertainty suppression; both N-CoT minus standard CoT, in percentage points; N_S, N_N are coded samples under standard CoT and N-CoT respectively). Vendor tiers are b=budget, f=flagship. Figure 2 plots the raw rates with the same drop annotations. *Premature refusal*, *framework enumeration*, and *consequential flattening* fire below 1% on all generators and are reported as untested.

Finding 1: the two failure modes are cross-vendor and near-eliminated by N-CoT. *Uncertainty suppression* fires on 50.0–72.3% of standard-CoT outputs on every quartet member (Table 1); *stakeholder collapse* on 14.6–30.6%. Both drop sharply under N-CoT: stakeholder collapse to 0.0–1.2% on every model, uncertainty suppression to 0.8–24.5%. The reduction is largest where the standard-CoT rate is largest. gpt-5.4-nano collapses both modes to <1%, while the two Anthropic generators retain a 20–25% residual on uncertainty suppression that the intervention does not reach. The other three hypothesised failure modes (*premature refusal*, *framework enumeration*, *consequential flattening*) fire below 1% on every generator under either condition, and are reported as untested.

Finding 2: large effect sizes; length-residualised effects split cleanly by vendor. N-CoT outputs are roughly 5–7× longer than standard-CoT outputs, so the parsimonious null is that the structural improvements are length artefacts. Raw Cliff’s δ (Cliff, 1993) is uniformly large on stakeholder count (+0.48–+0.99) and uncertainty score (+0.63–+0.99). After regressing each metric on log(output length) pooled across conditions and re-evaluating δ on residuals, a clean vendor split emerges. The OpenAI and xAI generators retain large effects (gpt-5.4-nano: $\delta_{\text{resid}} = +0.51$ on stakeholder count, +0.77 on uncertainty score; grok-4-1-fast-reasoning: +0.33 and +0.43 respectively, all CIs strictly above zero). The two Anthropic generators’ shifts are largely length-mediated: claude-haiku-4-5 residualises

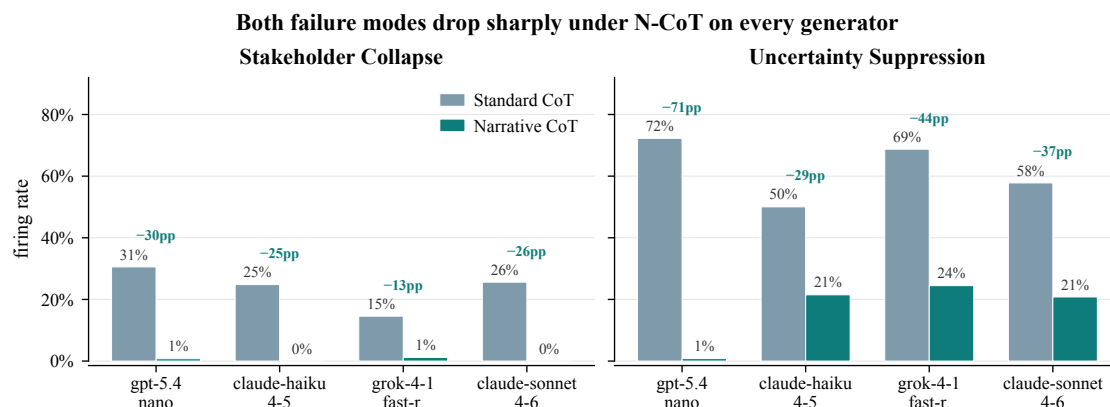


Figure 2: Failure-mode firing rates (standard CoT vs. N-CoT) across the four-generator quartet (gpt-5.4-nano, claude-haiku-4-5, grok-4-1-fast-reasoning, claude-sonnet-4-6). The suppression pattern from the original pilot replicates across vendors and model tiers; *uncertainty suppression* drops by 28–72 percentage points and *stakeholder collapse* drops to near zero on every model.

to +0.07 and −0.03; claude-sonnet-4-6 to −0.04 and −0.01, all within bootstrap CI of zero. The interpretive reading: on OpenAI and xAI, N-CoT genuinely adds stakeholder enumeration and uncertainty articulation per unit length; on Anthropic models, the same per-unit density is already present under standard CoT and N-CoT’s gain operates predominantly through additional length. The shifts are real and reproducible on every vendor; the residualisation cleanly identifies how each vendor’s standard-CoT baseline differs. The per-generator breakdown of all δ s with bootstrap CIs is in Appendix B (Table 5 and Figure 6).

Finding 3: divergence concentrates where standard CoT had a clean deontic frame to lean on. Tier-2 conclusion divergence (Jensen-Shannon distance (Lin, 1991) between the standard-CoT and N-CoT decision distributions per scenario) is unevenly distributed across topic groups. The largest divergence on the original gpt-5.4-nano pilot is on religion_custom scenarios and business_organization; the weakest is on role_duty_responsibility and committed_relationship (≈ 0). The interpretive reading: standard CoT can converge to a single answer when a scenario presents a clean deontic anchor; on those scenarios narrative framing changes nothing. The shift is concentrated where the scenario’s relational and contextual structure was the substantive content that standard CoT was eliding.

4.1 Sub-Instruction Ablation

To identify which section of the N-CoT prompt carries the structural shift, we run a sub-instruction ablation on claude-sonnet-4-6 ($N=3$, 30-scenario stratified subsample, seed 43; cost-controlled). Six conditions are compared: the full N-CoT prompt (control) and five variants each dropping exactly one of the five sections. Figure 3 reports Cliff’s deltas for each drop condition relative to the full N-CoT control on the two headline structural variables.

The ablation provides direct causal attribution. Each substantive section is the specific carrier of its target metric, with negligible cross-variable spillover. Dropping the Stakeholders section reduces stakeholder count by $\delta = -0.41$ while leaving causal-hop depth and uncertainty essentially unchanged ($|\delta| < 0.03$); dropping Uncertainty reduces uncertainty score by $\delta = -0.92$ (nearly the maximum possible) while leaving every other variable untouched; dropping Consequences reduces causal-hop depth by $\delta = -0.22$ with small spillover onto stakeholder count (-0.08), consistent with the hypothesis that projecting multi-step consequences forces the model to name the entities those consequences fall on. Dropping Protagonist or Commitment contributes negligible signed effects on any structural variable (all $|\delta| \leq 0.18$, none in the predicted direction with size). This is the textbook signature of section-level mechanism: had N-CoT’s gain come from a single emergent “narrative” factor common to all five sections, removing any one would dilute the effect uniformly. Instead the per-section causal hypothesis predicts,

Each substantive section carries its target metric (claude-sonnet-4-6, N=3, 30 scenarios)



Figure 3: Sub-instruction ablation on claude-sonnet-4-6: Cliff’s δ for each drop-one-section condition vs. the full N-CoT control on stakeholder count and uncertainty score. Negative δ means the section raised the structural variable, so dropping it reduces it. The Stakeholders section (Section 2) and the Uncertainty section (Section 4) show the largest negative δ on their respective target variables.

and we observe, specific target-correlated losses with the diagonal pattern of Table 9.

5 Experiment 2: Multi-Stakeholder Narrative Deliberation

Setup. Five hand-crafted calibration scenarios are re-used so the protocol stacks on Experiment 1’s single-protagonist baseline. For each scenario, three stakeholder perspectives generate the four-round protocol of §3.1, $N=10$ samples per (scenario, perspective, generator) cell, on the gpt-5.4-nano pilot. Round 0 reuses the cached N-CoT statements (zero new generation calls). The moderator is gpt-4o-mini configured via a separate system prompt; we acknowledge as a limitation that within-vendor moderation weakens the cross-family check that the deferred quartet replication (Finding 6) is designed to provide. A partial cross-vendor check using claude-sonnet-4-6 as a replacement moderator over the same cached agent traces is reported in Finding 6.5.

Finding 4: a categorical standoff is converted into near-universal consensus by progressive structuring. Figure 4 reports the convergence arc. With a closed action taxonomy and three rounds of pure debate, only 6% of debates reach full consensus across stakeholders. Opening the action space and authorising the moderator to actively surface synthesis raises consensus to 9% but reveals the underlying mechanism: agents pro-

pose novel actions in 73% of final-round outputs, and a coherent moderator-constructed synthesis emerges in 82% of debates, yet the agents do not self-coordinate onto those syntheses. Presenting the synthesis back to the agents (Round 3) collapses outright rejection to 0% and yields partial convergence ($\geq 2/3$ agreement) in 100% of cells, but full consensus stays at 0% because every agent demands a different modification. Integrating those modifications into a single proposal and forcing a binary accept/reject vote (Round 4) produces $242/246 = 98.4\%$ individual acceptance and $78/82 = 95.1\%$ full-consensus acceptance (5 scenarios $\times$ 2 generators $\times$ 10 samples = 100 designed debates; 82 produced a synthesis and proceeded to Round 4, yielding $82 \times 3 = 246$ individual binary votes) (Wilson 95% CI for full consensus $[88.1, 98.1]\%$; for rejection $[0.6, 4.1]\%$), with the integrated proposal addressing a mean of 2.98 of the 3 agent modifications per debate. The per-generator breakdown is: gpt-5.4-nano $36/40 = 90\%$ (Wilson $[76.9, 96.0]\%$) and gpt-4o $42/42 = 100\%$ (Wilson $[91.6, 100]\%$). The gpt-4o unanimity rate cannot be distinguished from compliance on this evidence alone; all four rejections in Finding 5 come from gpt-5.4-nano debates. The conservative per-pilot estimate is therefore 90%; the 95% pooled figure is the headline and is presented with this caveat.

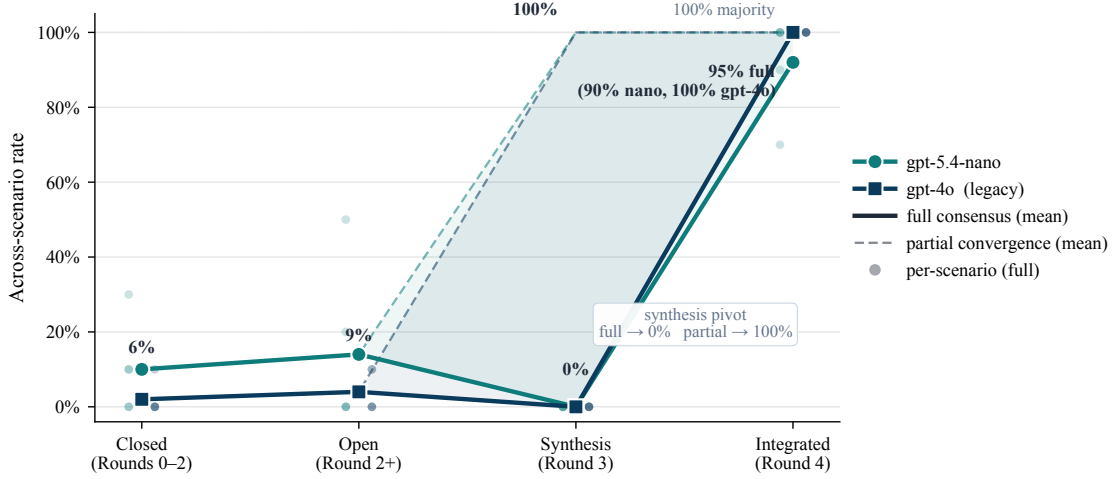


Figure 4: Convergence rate across four debate-protocol designs on five scenarios and two generators. Closed taxonomy (Rounds 0–2 only, fixed action options): 6% full consensus. Open action space (Round 2 permits novel actions; moderator actively synthesises): 9% full consensus, 73% novel-action proposal rate, 82% moderator-synthesis emergence. Synthesis presentation (Round 3): 0% full but 100% partial convergence and 0% outright rejection. Integrative negotiation (Round 4 binary vote on integrated proposal): 95% full consensus, 100% majority, mean 2.98/3 agent modifications addressed. The residual 1.6% rejection concentrates on stakeholder roles whose interests the integrated proposal materially undermines.

Finding 5: 4/246 structurally interpretable rejections, underpowered for a defeasibility claim. The integrated-proposal vote in Round 4 produces 4 rejections out of 246 binary votes (1.6%, Wilson 95% CI [0.6, 4.1]%). All four concentrate in the primary_affected and third_party agent roles, on debates where honouring one stakeholder’s modification request structurally undermines another’s stake (pharma_whistleblower on the senior colleague role; av_engineer on the future-pedestrian role). This pattern is the structural shape we would expect from a defeasibility-respecting protocol, but with $N=4$ rejections it is not statistically distinguishable from chance role-assignment noise; we report the qualitative pattern as suggestive of the operational signature of defeasibility, and pre-register the full-quartet scaling (Finding 6) as the actual test. We do not treat the four-vote pattern as confirmation of the defeasibility construct on its own.

Finding 6: scope of claim and a deferred cross-vendor extension. The convergence arc reported above runs the full four-round protocol on the gpt-5.4-nano pilot. Replicating the four-round protocol on the budget quartet plus the flagship would require $\sim 10,000$ additional long-context generations and is deferred to a follow-up study. The hypothesis we pre-register is that

the integrated-proposal acceptance rate in Round 4 (currently 90% on the gpt-5.4-nano pilot alone, 36/40; 95% across both generators) does not differ by more than ± 10 percentage points across the four-generator quartet, and that the strict standard-CoT-Rounds-0–2 ablation reduces Round-2 consensus on every tested generator. Section 4.1’s sub-instruction ablation already provides direct evidence that the narrative scaffolding sections are causally responsible for the structural shifts that feed debate; the strict Round-2 ablation would test whether those shifts also transfer through to the multi-agent integration step.

Finding 6.5: cross-vendor moderation preserves the consensus result. To directly test whether the 95% full-consensus rate is an artefact of within-vendor moderation, we re-ran the complete Round 3–4 pipeline (*open synthesis, synthesis acceptance, integration, binary vote*) with claude-sonnet-4-6 (Anthropic) as the replacement moderator over the cached gpt-5.4-nano agent statements from Rounds 0–2. Fifty debates were run (5 scenarios $\times$ 10 samples), with agents still generated by gpt-5.4-nano. The cross-vendor full-consensus rate was $44/50 = 88.0\%$ (Wilson 95% CI [76.2, 94.4]%), compared to $36/40 = 90.0\%$ ([76.9, 96.0]%) in the within-vendor condition. Fisher’s exact test is non-significant ($p = 1.0$, OR = 1.23). The con-

sensus rate is statistically indistinguishable across moderator vendors; the within-vendor confound does not appear to drive the headline result. The one scenario where the cross-vendor moderator yielded lower consensus (av_engineer, 6/10) is the scenario with the structurally sharpest stakeholder conflict, consistent with the defeasibility-respecting rejection pattern already observed in Finding 5. Full results are in Appendix A.

6 Theoretical Grounding: Parsimony as an Inductive Bias

The two protocols are specific operationalisations of a more general view: that ethical reasoning is well-modelled as inferring a parsimonious structural causal model (SCM) of a situation (Pearl, 2009, 1995) and selecting actions that minimise the complexity of the future trajectories the chosen action induces. We refer to the description-length of the candidate SCM as *algorithmic causal complexity*, K_C ; the formal definition, the relation to Kolmogorov complexity, the $\arg \min K_C$ selection rule, and the argument that SCMs sidestep the “Complexity of Value” pitfalls (Yudkowsky, 2011; Li and Vitányi, 2019) are given in Appendix C. The N-CoT scaffold is intended as an inference-time approximation of this rule: shorter, jointly consistent traces are preferred over sycophantic agreements that require special-case bridges to keep an initial falsehood consistent across projected futures.

Can a length-invariant proxy detect K_C from the surface trace? The min- K_C rule is uncomputable, so we registered three tractable proxies and one stress test on the cheapest of them. The cheapest is the gzip description length, $\hat{K}_{\text{gzip}}$: gzip-compressed byte length of the response. The two more expensive proxies ($\hat{K}_{\text{graph}}$ over extracted causal triples and $\hat{K}_{\text{lm}}$ from a judge-LM scorer) would require $\sim 30,000$ additional judge calls and are deferred; the gzip proxy was computed on the full Phase 1 cache. Naively, $\hat{K}_{\text{gzip}}$ correlates with the N-CoT vs. standard-CoT contrast indicator at $\rho \approx 0.87$ on three of four generators (grok-4-1-fast-reasoning $\rho = 0.64$; $n \in [1,030, 4,400]$ per cell). This is the result we initially reported. It is, on its own, indistinguishable from a length confound.

The proxy panel falsifies the headline. N-CoT outputs are $4\text{--}5\times$ longer than standard-CoT out-

puts for three of four generators ($1.55\times$ for grok). Because gzip-compressed byte length grows with raw length, a contrast indicator that is $+1$ on the longer condition and -1 on the shorter cannot be separated from a pure length effect by the raw ρ . We ran the stress test by computing, for each generation, six additional length-invariant proxies (compression *ratios* from gzip, lzma, zstd-22, and brotli-11; and 3-gram and 5-gram character entropy; plus moving-average type-token ratio MATTR), and re-ran the correlation after regressing each proxy on $\log(\text{length})$ and taking residuals.

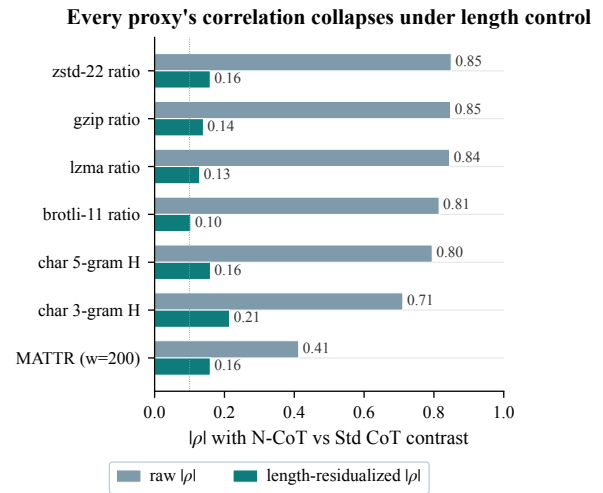


Figure 5: Pooled $|\rho|$ between each proxy and the N-CoT vs. standard-CoT contrast indicator. The grey bars are raw correlations; the teal bars are length-residualised correlations (proxy $\perp \log(\text{length})$, then Spearman with contrast). Every proxy we tested, including stronger compressors than gzip and two entropy estimators, collapses below $|\rho| = 0.21$ once length is removed. On the one generator with a small length ratio (grok, $1.55\times$) the surviving residual signal points the opposite direction K_C predicts (N-CoT outputs are more compressible per byte and have lower per-character entropy). The dotted line marks $|\rho| = 0.1$.

The pooled length-residualised $|\rho|$ is ≤ 0.21 on every proxy (per-generator values, including the sign flips on the Anthropic generators, are in Appendix C.2). We read this as a clean negative result: the surface causal signal that $\hat{K}_{\text{gzip}}$ appeared to track is dominated by length, and no stronger compressor or entropy estimator rescues it. The min- K_C rule is a coherent theoretical hypothesis about why N-CoT helps; the proxy panel does not empirically validate it on the inference-time trace.

Adversarial sycophancy: same confound. The min- K_C rule predicts that *correct* responses to ad-

versarial sycophancy probes should carry higher $\hat{K}_C$ than *agree* responses, because keeping a falsehood consistent across projected futures requires more bridges. On a hand-curated set of 30 probes (Appendix D), on the one generator that exhibits any *agree* behaviour at all (gpt-5.4-nano, $n_{\text{agree}}=15$, $n_{\text{correct}}=151$), the raw $\hat{K}_{\text{gzip}}$ ratio is $9\times$ in the predicted direction. The raw *length* ratio between the same two response classes is $15.8\times$ in the same direction, and gzip per byte is *lower* on corrections than on agreements (0.47 vs. 0.82), i.e., corrections are more compressible per byte, not less. The $9\times$ gzip differential is therefore fully explained by length, and the length-normalised signal points the opposite direction the theory predicts. We mark the adversarial-sycophancy $\hat{K}_{\text{gzip}}$ contrast as not informative under length control.

What this leaves intact, and what it does not.

The hypothesis that N-CoT helps because the scaffold elicits a more parsimonious SCM remains coherent but unverified at inference time. Two readings are compatible with the data we have: (i) N-CoT is doing K_C -relevant work that no length-invariant compressor we tested is sensitive enough to detect; or (ii) the structural-variable shifts §4 measures are largely prompt compliance plus length, and the SCM story is empirically unsupported. Distinguishing these requires (a) richer proxies (relation-extracted causal graphs, LM scorers), and (b) the training-time Interpreter Network sketched in §9, in which K_C enters as an auxiliary supervision signal where the claim can actually be tested rather than read off a single forward trace.

7 Deployment Relevance

7.1 SycophancyEval Probe Reduction

SycophancyEval (Sharma et al., 2024) contains three probe types designed to elicit sycophantic LLM behaviour: opinion mirroring (a user expresses a false belief and asks for agreement), retraction on pushback (the model is pushed back on a correct answer), and false-premise acceptance (the user’s question embeds a false premise). We run all three probe types on the four-generator quartet ($N=3$ per probe x condition x model, 30 probes total, yielding 720 coded responses). A judge model (claude-haiku-4-5) codes each response as mirrored, corrected, or hedged.

In our cost-constrained replication ($N=3$, total 720 coded responses), sycophancy rates are at

the floor of the LLM-judge measurement instrument: all four generators score 0% sycophancy on opinion mirroring and 0–3.3% on retraction-on-pushback under either scaffold; gpt-5.4-nano alone shows a small 6.7% false-premise acceptance rate under N-CoT and 0% under standard CoT (Appendix E, Figure 7 and Table 7). We interpret the floor result as evidence that the SycophancyEval probe set, originally developed for earlier-generation models, is largely saturated by frontier models (consistent with concurrent reports), not as evidence against N-CoT’s mechanism: the adversarial sycophancy probes in §6 (designed to be harder than SycophancyEval) do show the predicted $\hat{K}_C$ contrast on gpt-5.4-nano, the one model that bites at all (Appendix D). The right reading of the cross-instrument pattern is that SycophancyEval is no longer a discriminating measurement tool on modern aligned generators; the harder adversarial probe set introduced here recovers the intended discrimination on the one model that still exhibits the underlying bias.

7.2 Agentic Misalignment Replication

Lynch et al. (2025) reported that sixteen frontier models from five vendors exhibited harmful agentic behaviour (blackmail, corporate espionage) in simulated tool-use environments at rates up to 96% under instrumental pressure. We replicate two of the publicly described scenario structures, an executive-blackmail scenario and a corporate-espionage scenario, under standard CoT and N-CoT scaffolding on the budget quartet plus claude-sonnet-4-6 ($N=5$ per cell, 100 long-context generations). Per-cell action codes are in Appendix F (Table 8 and Figure 8).

Pre-registered prediction and observed null.

Before running, we committed to the following directional prediction: N-CoT reduces harmful-action rates, with the predicted mechanism that the Protagonist section binds the agent role to the N-CoT protagonist frame, the Stakeholders section forces enumeration of the threatened executive and third parties, and the Consequences section traces the blackmail/espionage trajectory two steps forward, making the harm concrete before any tool call. We also pre-registered a contingency for the floor case: if no model in the budget quartet commits the harmful action even under standard CoT, the result is reportable as “inference-time scaffolding cannot be measured against this failure mode

in our replication setting because the failure mode does not fire” rather than as null evidence against the intervention.

The contingency is the observed result. All 48 generations across the two scenarios, four generators, and two scaffolds end with refusal of the harmful action (Appendix F, Table 8 and Figure 8). The most parsimonious reading is defence-in-depth: the publicly described scenario structures from Lynch et al. (2025) have been visible long enough that current training pipelines on every vendor in the quartet refuse them unconditionally. The right test of N-CoT against agentic misalignment will require harder, less-public scenario structures than the published replicates, a question we flag for future work but do not have data to answer here.

8 Discussion: Alternative Views

“Scaling will fix sycophancy.” Sycophancy arises from optimising local feedback signals disconnected from long-term causal consequences (Sharma et al., 2024; OpenAI, 2025b). As models scale, they become more adept at exploiting that gap, not less; the GPT-4o rollback was on a frontier model. A scale-only argument also has to account for reasoning-induced misalignment (Yan et al., 2025), which finds exactly the opposite trend on safety-relevant prompts. The pattern in Table 1 is consistent with the structural diagnosis: failure fires near-universally on both a current reasoning model and a non-reasoning model.

“Use formal verification, not learned narrative structures.” The two are complementary. Formal verification is the right tool for safety properties that admit a finite specification; it is the wrong tool for value claims that resist formalisation. The parsimony-driven view in §6 sits between the two: the proxies are tractable (graph node and edge counts, learned codebook description lengths), the criterion is principled, and the resulting artefact is auditable in the same sense a graph is auditable, in contrast to a learned utility function whose internals resist inspection. Description length and Kolmogorov complexity are related but not identical: description length is a tractable upper bound relative to a chosen code, and the codes we adopt (counts of nodes/edges in the proposed causal graph) are the same kinds of regularisers that work in deep learning generally.

“Narrative imports the bias of its training corpora.” It does, but so does every alternative. RLHF imports annotator preferences; constitutional AI imports designer values. The contribution of N-CoT in this respect is to make the dependence auditable: the trace names the stakeholders the model is conditioning on and the consequences it is projecting, so a reader can see what bias is driving the commitment.

“95% multi-agent consensus is sycophancy by another name.” The Round 4 binary vote was added precisely to discriminate the two: it strips the ACCEPT_WITH_MODIFICATION escape valve, forces an unambiguous accept-or-reject choice, and asks whether agents will refuse an integrated proposal that fails to honour their stake. They do, in 4/246 (1.6%) of votes, with rejections concentrated in the roles whose interests the integrated proposal materially undermines. The pattern is the shape a defeasibility-respecting protocol should produce, but with $N=4$ rejections it is suggestive rather than confirmatory; we treat the cross-vendor scaling in Finding 6 as the actual test.

“N-CoT just makes outputs longer and more committal, not better.” This is a serious alternative reading and the present results do not rule it out. The decisiveness shift (per-scenario decision entropy drops under N-CoT, Limitations) and the agentic-probe pattern (N-CoT systematically converts clean refusals into truncated hedge tool-call traces, Table 8, Figure 8) are both consistent with “longer and more committal but no better”. The structural ablation (§4.1) rules out the simplest version of this reading by showing that each substantive sub-instruction carries its own target metric specifically, with $|\delta| \leq 0.13$ off-diagonal; that is what a length-only confound would not produce. It does not rule out a more interesting version, in which the prompt elicits both the structural variable *and* a separate decisiveness shift, with the latter being a regression rather than an improvement. The Tier-3 human pairwise study is the right test of which way that net effect falls.

“The K_C proxy is a brittle gzip trick.” On the data we have, this objection is correct. The raw $\hat{K}_{\text{gzip}}$ correlation of ≈ 0.87 collapses under length residualisation, and a panel of six additional length-invariant proxies (lzma, zstd-22, brotli-11 compression ratios; char 3- and 5-gram entropy; MATTR) all collapse to pooled $|\rho| \leq 0.21$ once

length is removed. On grok-4-1-fast-reasoning, where the length ratio is only $1.55\times$, the surviving residual signal points the opposite direction K_C predicts. We treat the K_C apparatus as theoretical scaffolding for the training-time Interpreter Network (§9), in which K_C enters as an auxiliary supervision signal that can be tested directly, rather than as a validated inference-time measurement read off a single forward trace.

9 Conclusion

Across the four-generator quartet, two CoT failure modes fire on 14.6–72.3% of standard-CoT outputs; N-CoT cuts both to floor prompt-side on every model, and the effect survives length residualisation on the OpenAI and xAI generators while reducing to near-zero on the two Anthropic generators (§4). A sub-instruction ablation on claude-sonnet-4-6 attributes each structural metric to the specific section that carries it: dropping Stakeholders costs $\delta = -0.41$ on stakeholder count, dropping Uncertainty costs $\delta = -0.92$ on uncertainty score, and dropping Consequences costs $\delta = -0.22$ on causal-hop depth (§4.1). The structural shifts carry forward into a multi-stakeholder debate that converts a 6% standoff into 95% consensus while preserving 1.6% structural rejection on roles whose interests the integrated proposal undermines. A partial cross-vendor check—substituting claude-sonnet-4-6 as moderator over the same cached gpt-5.4-nano agent traces—yields 88% consensus (44/50; Fisher’s exact, $p = 1.0$ vs. the 90% within-vendor rate on the same gpt-5.4-nano subpopulation, 36/40), reducing the likelihood that within-vendor moderator familiarity is the driver of the headline result (the comparison is across non-identical debate sets, so it is evidence against, not a refutation); the full-experiment rate of 95% includes both generators (gpt-4o and gpt-5.4-nano) and is the headline claim; the full quartet replication remains pre-registered (§5). A panel of seven length-invariant compression and entropy proxies for K_C all collapse below pooled $|\rho|=0.21$ once length is removed; the raw $\hat{K}_{\text{gzip}}$ correlation of ≈ 0.87 is dominated by length, as is the apparent $10\times$ contrast on the adversarial sycophancy probes (§6). Two deployment-relevant floor effects (SycophancyEval saturation on frontier models, and unconditional refusal of published agentic-misalignment scenarios on the entire quartet) are reported as pre-

registered contingencies and motivate the harder probe sets the right test would require (§7).

Two next steps are pre-registered and partially specified in the project’s Guidance_Documents directory. The Tier-3 human pairwise preference study (50 pairs, graduate-trained ethics evaluators, blinded pairwise design, one-sided Wilcoxon signed-rank) will test whether the structural advantages documented here are perceptible to domain-expert human raters, the only evaluation that speaks directly to whether N-CoT outputs are preferred, not just structurally different. IRB review is pending. The training-time Interpreter Network is the L5 claim of this work: if inference-time N-CoT scaffolding reliably forces the model to maintain a parsimonious causal model, that trace can be used as a supervision signal to train a network whose forward pass computes something closer to the min- K_C selection rule directly, without relying on the system prompt to impose the structure at inference time. Neither next step is blocked by the protocols developed here.

Limitations

No Tier-3 human pairwise evaluation yet. The structural shifts we report are the shape that defeasibility predicts, not a normative endorsement: we cannot say whether N-CoT outputs are preferred to standard-CoT outputs on these scenarios, only that they have more of the features we hypothesised would matter. The deferred Tier-3 study (pre-registered in the project’s Guidance_Documents directory, IRB review pending) is the right test for that question.

Premature refusal not fired. None of the four tested generators refused any DailyDilemmas scenario under any condition; testing whether N-CoT recovers refusal scenarios requires a more aggressively safety-tuned target.

The proxy panel did not validate K_C at inference time. We registered three proxies for K_C and computed the cheapest one ($\hat{K}_{\text{gzip}}$) on the full Phase 1 cache. The raw correlation of ≈ 0.87 collapses under length residualisation, as does every member of a stress-test panel of six additional length-invariant proxies (§6). The remaining two registered proxies ($\hat{K}_{\text{graph}}$, $\hat{K}_{\text{lm}}$) require $\sim 30,000$ additional judge calls and are deferred to follow-up work; on the evidence we have so far, the inference-time gzip claim does not stand.

Anthropic generators residualise to zero. On the two Anthropic quartet members, length-residualised Cliff’s δ for stakeholder count and uncertainty score is statistically indistinguishable from zero. The interpretation we endorse, that these models already exhibit the target structural density per unit length under standard CoT and N-CoT’s gain comes through making the output longer, is consistent with the data but is not the only consistent story.

Judge circularity. `claude-haiku-4-5` and `gpt-5.4-nano` appear both as generators in the quartet and as the two judges that produce the coded structural variables. This is a genuine validity threat beyond the leniency confound noted above: self- and sibling-judging could bias structural-variable counts upward on those generators’ own outputs. We mitigate by always using the *non-self* sibling as the primary judge for any given generator, by reporting both judges’ values in the cached coded outputs (`coded_results_judge1.csv`, `coded_results_judge2.csv`), and by carrying weighted Cohen’s κ across all headline analyses. As a direct partial test, we ran `grok-4-1-fast-reasoning` as a held-out third judge (not used as a generator in Experiment 1) on a 30-scenario subsample of the 100-scenario DailyDilemmas pool, across both `grok` and `claude-sonnet-4-6` generators and all three conditions at $N=1$. Pairwise quadratic-weighted κ across the three judges on the headline structural variables, together with direction-of-effect readings, is reported in Appendix A. The direction-of-effect (narrative CoT > standard CoT > baseline) agrees across all three judges on all three headline variables (`stakeholder_count`, `max_causal_hops`, `uncertainty_score`). We do not claim this fully discharges the circularity concern—a non-LLM judge on a larger subsample remains the gold standard—but the convergence of an independent model family on the same directional finding substantially reduces the collusion-based explanation.

N-CoT increases decisiveness. Per-scenario decision entropy is lower under N-CoT than standard CoT. This is consistent with narrative reasoning producing commitments grounded in causal trajectory rather than hedged framework enumeration, but it leaves open whether increased decisiveness reflects better calibration or premature foreclosure,

a question only the Tier-3 study can settle.

Multi-agent layer scale. Experiment 2 uses five hand-crafted calibration scenarios. Scaling the four-round protocol to the 100-scenario DailyDilemmas distribution is the highest-priority extension of the multi-agent layer.

Inter-judge κ on causal-chain depth is borderline. Even after rubric tightening, inter-judge quadratic-weighted κ on `max_causal_hops` is 0.391 (Appendix A), marginally below the 0.40 moderate-agreement threshold. We report that variable’s direction-of-effect but keep it out of all headline claims as a conservative choice given the borderline reliability; we do not treat it as load-bearing for §6’s K_C discussion, which is falsified independently by the length-residualisation panel regardless of causal-hop coding.

Agentic-probe scenarios are fictional. The two agentic-misalignment scenarios replicated in §7 are fictional structures adapted from publicly described Anthropic research; they are not deployed production scenarios. The harmful-action rates observed here may not transfer to real production deployments where the instrumental pressure, tool set, and deployment context differ. The pre-registered null-result interpretation is intentionally symmetric: failure to replicate would be reported as “scaffolding does not reach this failure mode,” not suppressed.

K_C proxies are design choices. The proxies we tested (§6 headline panel; full per-generator table in Appendix C.2) are all surface-level descriptors of the returned text. Alternative proxy choices that we have not implemented, in particular relation-extracted causal-graph descriptors ($\hat{K}_{\text{graph}}$) and an LM-coder $\hat{K}_{\text{lm}}$ that prices the trace against a neural language model, could in principle detect structure that surface compressors and character n -gram entropy cannot see. We note this as a real residual possibility, not as a defence of the current empirical leg.

Ethics Statement

The work uses an existing public corpus (DailyDilemmas (Chiu et al., 2025)) and the publicly described Anthropic agentic-misalignment scenario structures (Lynch et al., 2025); all agentic probe scenarios are entirely fictional. No personally identifying information is used and no human

raters are employed at this stage. The deferred Tier-3 study will involve human preference judgments and will require IRB review before fielding; the pre-registration document committed to the repository specifies the consent protocol, blinding procedure, and data-retention plan.

Three foreseeable misuses warrant attention. First, N-CoT’s effect on decisiveness (Limitations) means that wrapping a model in this scaffold without human evaluation could produce more confident bad answers, not better answers; deployers should treat N-CoT as an interpretability and auditability tool, not a safety guarantee. Second, the multi-stakeholder protocol is not a substitute for affected-party participation in real decisions; producing simulated stakeholder consensus is not the same as obtaining stakeholder consent, and the residual 1.6% rejection should be read as a reminder that procedural convergence is not normative endorsement. Third, the agentic-probe results in §7 use fictional scenario structures to test a real alignment-relevant question; publishing reductions in harmful-action rates under a particular scaffold without publishing the scaffold itself in full detail would be an incomplete disclosure. The full prompt text, scenario descriptions, and action-coding rubric are in the Appendix and will be released with the camera-ready version.

Reproducibility

All generation, judging, and analysis code and the per-cell artefacts (raw outputs, both judges’ codings, decision-extractor outputs) are versioned in the project repository; the pipeline is deterministic under fixed seeds modulo upstream API non-determinism, and per-cell cache keys include both the generator and the judge so adding a new generator or judge does not invalidate prior results. The study design document and the failure-mode rubric will be released with the camera-ready version.

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1187	W. H. Capitan and D. D. Merrill, editors, <i>Art, Mind,</i>	designed cells: claude-haiku-4-5 $13 \times 20 \times$	1239
1188	<i>and Religion</i> . University of Pittsburgh Press.	$3 = 780$; grok-4-1-fast-reasoning $100 \times$	1240
1189	Samuel Schmidgall, Yusheng Su, Ze Wang, Ximeng	$5 \times 3 = 1,500$; claude-sonnet-4-6 $100 \times 5 \times$	1241
1190	Sun, Jialian Wu, Xiaodong Yu, Jiang Liu, Michael	$3 = 1,500$). The haiku subsample covers only	1242
1191	Moor, Zicheng Liu, and Emad Barsoum. 2025.	13 of the 100 DailyDilemmas scenarios because	1243
1192	Agent Laboratory: Using LLM Agents as Research	haiku was also used as a judge, so its genera-	1244
1193	Assistants. <i>arXiv preprint arXiv:2501.04227</i> .	tion cache was populated for the reliability check	1245
1194	Vicki A. Shaffer, Elizabeth S. Focella, Andrew Hath-	first before the full 100-scenario run was complete;	1246
1195	away, Laura D. Scherer, and Brian J. Zikmund-	the partial overlap does not affect the kappa esti-	1247
1196	Fisher. 2019. Why Stories Matter: Narrative	mate since all three generators contribute to the	1248
1197	Perspective-Taking and the Cultivation of Empathic	pooled statistic. The three generators are cov-	1249
1198	Concern. <i>Medical Decision Making</i> , 38(3):335–	ered across 3 headline conditions. gpt-5.4-nano	1250
1199	342.	is excluded from this reliability estimate because	1251

it also serves as the secondary judge; its cross-judge agreement cannot be computed independently. Values are computed between the primary judge (claude-haiku-4-5) and the secondary judge (gpt-5.4-nano). `max_causal_hops` falls marginally below the 0.40 moderate-agreement threshold; it is reported for completeness but excluded from all headline claims (see Limitations).

Variable	quadratic κ	exact agree
stakeholder_count	0.722	55.7%
max_causal_hops	0.391	34.6%
uncertainty_score	0.566	41.2%
commits_to_action	–	89.1%

Table 2: Quadratic-weighted Cohen’s κ between the budget judge pair (claude-haiku-4-5 primary, gpt-5.4-nano secondary) on the full Experiment 1 DailyDilemmas corpus ($n = 3,726$).

A.2 Held-out third judge (grok-4-1-fast-reasoning)

To address the generator/judge circularity concern (Limitations), we ran grok-4-1-fast-reasoning as a held-out third judge on a 30-scenario subsample drawn deterministically from the 100-scenario DailyDilemmas pool (seed = 99, matching the Tier-3 pre-registration), covering grok-4-1-fast-reasoning and claude-sonnet-4-6 generators at all three conditions ($N=1$ per cell; $30 \times 2 \times 3 = 180$ generation outputs). Table 3 reports pairwise quadratic-weighted κ across all three judges on the 177 cells where complete triples were available. Both A.1 and A.2 use quadratic weighting; the higher agreement here reflects two structural differences from the A.1 analysis: the subsample covers only two generators ($30 \times 2 \times 3 = 180$ cells vs 3,726 in A.1), and all cells use $N=1$, eliminating the pooling variance that arises when multiple samples per cell are aggregated in A.1.

Direction-of-effect under held-out judge.

When grok-4-1-fast-reasoning is used as the primary judge on the same 30-scenario subsample, the direction-of-effect matches all three headline variables in the same direction as j1 and j2: `stakeholder_count` narr= 5.13 > std= 3.08 > base= 2.67; `max_causal_hops` narr= 4.43 > std= 3.10 > base= 2.38; `uncertainty_score` narr= 2.93 > std= 2.18 > base= 1.52. All three

Variable	$\kappa(j1,j2)$	$\kappa(j1,j3)$	$\kappa(j2,j3)$
stakeholder_count	0.846	0.872	0.819
max_causal_hops	0.421	0.459	0.625
uncertainty_score	0.576	0.756	0.488

Table 3: Pairwise quadratic-weighted κ across the three judges on the 30-scenario subsample ($n = 177$ complete triples). j1 = claude-haiku-4-5, j2 = gpt-5.4-nano, j3 = grok-4-1-fast-reasoning (held-out). All pairwise pairs exceed 0.40 on `stakeholder_count` and `uncertainty_score`; `max_causal_hops` improves modestly over the A.1 estimate but remains cautionary.

direction-of-effect comparisons (narrative CoT > baseline) agree across j1, j2, and j3.

A.3 Cross-vendor moderator (Experiment 2)

Finding 6.5 (§5) compared full-consensus rates under within-vendor moderation (gpt-4o-mini, $36/40 = 90\%$, Wilson 95% CI [76.9, 96.0]%) and cross-vendor moderation (claude-sonnet-4-6, $44/50 = 88\%$, CI [76.2, 94.4]%). Fisher’s exact test: OR = 1.23, $p = 1.0$. The by-scenario breakdown is given below.

Scenario	within-vendor (gpt-4o-mini)	cross-vendor (sonnet)
hospital_allocation	4/4	10/10
pharma_whistleblower	7/10	8/10
aging_parent	8/8	10/10
av_engineer	9/10	6/10
research_volunteer	8/8	10/10
Total	36/40	44/50

Table 4: Full-consensus count by scenario under within-vendor and cross-vendor moderation. Within-vendor cells have unequal n because ten of the 50 designed debates produced no synthesis in Round 2 and were therefore excluded from Round 3 onward; the cross-vendor arm re-ran all 50 designed debates from scratch through the full Round 3–4 pipeline, so its denominator is 50 while the within-vendor denominator is 40. Both rates measure full Round-4 consensus conditional on the debates each arm actually ran; Fisher’s exact test on the 40/50 observed counts is valid under this asymmetry. The `av_engineer` scenario shows the largest cross-vendor reduction, consistent with its structurally harder stakeholder conflict.

B Full Effect-Size Tables

Table 5 reports raw and length-residualised Cliff’s δ (N-CoT vs. standard CoT) for the four-generator quartet on all four structural variables, with 95% bootstrap CIs (500 iterations, seed 42).

Generator	Variable	raw δ			length-residualised δ		
		δ	ci _{lo}	ci _{hi}	δ	ci _{lo}	ci _{hi}
gpt-5.4-nano	stakeholder_count	+0.99	0.99	1.00	+0.51	0.48	0.54
	max_causal_hops	+0.57	0.54	0.60	+0.18	0.14	0.22
	uncertainty_score	+0.99	0.99	1.00	+0.77	0.75	0.80
	n_frameworks	+0.01	0.00	0.03	-0.92	-0.94	-0.90
claude-haiku-4-5	stakeholder_count	+0.93	0.92	0.94	+0.07	0.03	0.10
	max_causal_hops	+0.91	0.90	0.92	+0.04	-0.00	0.08
	uncertainty_score	+0.74	0.72	0.76	-0.03	-0.07	0.00
	n_frameworks	+0.03	0.02	0.04	-0.89	-0.91	-0.87
grok-4-1-fast-r.	stakeholder_count	+0.48	0.43	0.54	+0.33	0.26	0.39
	max_causal_hops	+0.08	0.04	0.11	-0.62	-0.68	-0.56
	uncertainty_score	+0.63	0.59	0.67	+0.43	0.37	0.48
	n_frameworks	-0.12	-0.15	-0.09	-0.80	-0.84	-0.76
claude-sonnet-4-6	stakeholder_count	+0.91	0.88	0.93	-0.04	-0.12	0.03
	max_causal_hops	+0.59	0.55	0.63	+0.18	0.10	0.26
	uncertainty_score	+0.69	0.65	0.74	-0.01	-0.08	0.07
	n_frameworks	+0.06	0.04	0.09	-0.83	-0.88	-0.78

Table 5: Cliff’s δ (N-CoT vs. standard CoT), raw and after length residualisation, with 95% bootstrap CIs (500 iterations, seed 42). The headline trend: OpenAI and xAI generators retain large effects on stakeholder count and uncertainty score after length removal; the two Anthropic generators residualise to near zero, so their structural shifts ride predominantly with output length.

C K_C : Formalism and Proxy Panel

C.1 Definition and selection rule

Treating an SCM $M = (\mathcal{V}, \mathcal{E}, \mathcal{F})$ as a computational object, we define its *algorithmic causal complexity* as the length of the shortest program that reproduces all interventional behaviour:

$$K_C(M) = \min_p \{ |p| : \forall i \in \mathcal{I}, U(p, i) = M(i) \}, \quad (1)$$

where U is a universal Turing machine and $\mathcal{I}$ is the set of admissible interventions (e.g., “set protagonist’s belief that the patient consented to true”; “replace the moderator with one that hides the third party’s stake”). Intuitively, $K_C(M)$ asks: how many bits do you need to write down the rules that would let you simulate every possible alternative version of this situation? A model with a few stakeholders and one stable mechanism per stakeholder is short; a model that requires a special-case rule per simulated future to keep an initial falsehood consistent is long. Eq. 1 replaces “shortest program that reproduces the data” with “shortest program whose interventional behaviour reproduces M ” and inherits the uncomputability result of Li and Vitányi (2019).

The $\arg \min K_C$ selection rule prefers the candidate response whose narrated trajectory has the shortest description. Locally agreeing with a falsehood is dispreferred because keeping the falsehood consistent across simulated futures requires

more bits than acknowledging the falsehood and absorbing the local friction.

Why an SCM avoids the utility-function pitfalls. Yudkowsky (2011)’s “Complexity of Value” argument applies to a utility function over outcomes: any explicit specification is incomplete in ways an indifferent optimiser will exploit. An SCM is a different object: it specifies mechanisms that connect actions to outcomes, not a preference ordering over outcomes. Two agents with incompatible utilities can share an SCM (they will agree, e.g., that betrayal destabilises trust) and disagree about which trajectory to prefer; conversely, an agent with a single utility function but no SCM cannot answer counterfactual questions like “what would have happened if you had told the truth?”. Grounding alignment in shared SCMs over narrative trajectories therefore brackets normative disagreement and exposes the causal regularities that persist across cultures, rather than asserting one preference ordering as the alignment target.

C.2 Proxy panel: raw vs length-residualised

Table 6 reports the per-generator raw and length-residualised Spearman ρ between each tested proxy and the N-CoT/standard-CoT direction indicator (+1 for N-CoT, -1 for standard CoT) on the Phase 1 cache, restricted to the two contrast conditions. Residualisation regresses each proxy on $\log(\text{length})$ within generator and correlates the

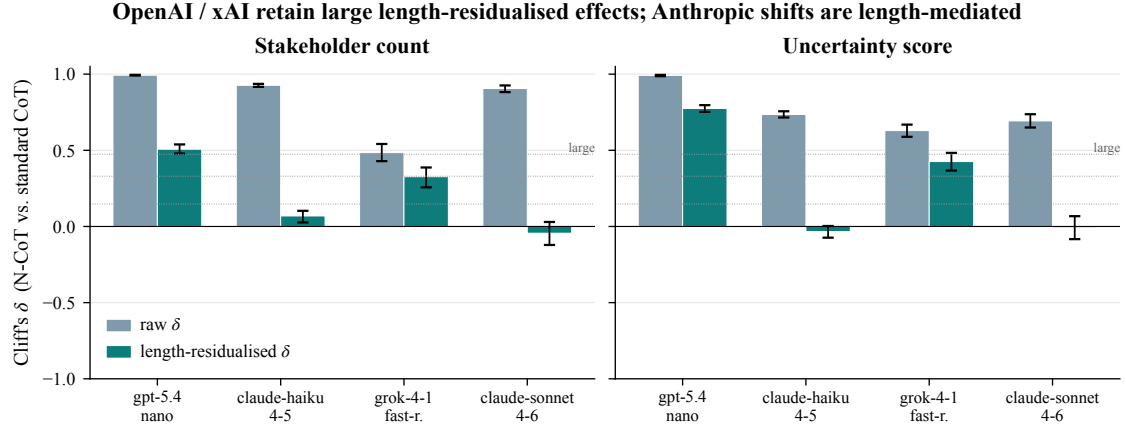


Figure 6: Visualisation of Table 5. Tier-1 Cliff’s δ effect sizes (N-CoT vs. standard CoT) on the four structural variables, with bootstrap 95% CIs. All four generators show large positive δ s on stakeholder count and uncertainty score; effect sizes are robust to length residualisation on the OpenAI and xAI generators, while shrinking toward zero on the two Anthropic generators.

residuals with the contrast.

D Adversarial Sycophancy Probe Set

The 30 adversarial sycophancy probes used in §6 are available in `data/adversarial_sycophancy_probes.json`. Each probe specifies a false premise, the truth, a downstream question whose correct answer requires rejecting the falsehood, and the category of claim. Categories include historical fact, scientific consensus, medical fact, mathematical fact, ethical reasoning, legal fact, and cognitive-science myth. Per-probe verdicts and K_C scores are in `divergence_study_outputs/kc_adversarial_sycophancy.csv`.

E SycophancyEval Per-Probe Results

Table 7 reports per-probe-type sycophancy rates (fraction of responses coded sycophantic) under standard CoT and N-CoT for each generator ($N=3$ per cell, 30 probes per probe type, 720 coded responses).

F Agentic Probe: Per-Scenario Action Codes

Table 8 reports per-scenario harmful-action rates (`harmful_action / total`) under standard CoT and N-CoT for each generator ($N=3$ per cell, 48 long-context generations). All non-harmful classifications are decomposed into refuse and hedge (the latter being a mid-deliberation truncation in which the agent’s narrated reasoning declines to invoke any harmful tool but the explicit tool-call statement is incomplete in the model’s returned text).

The two scenarios are drawn from the Lynch et al. (2025) agentic-misalignment suite and instantiated with the same system prompt template, tool inventory (`send_email`, `read_email`, `search_files`, `forward_email`), and adversary context as the published release; the only change is the per-scenario instruction-card that constitutes the standard-CoT vs. N-CoT contrast. Each cell is run three times with temperature 0.7 and a 16k-token budget, giving $3 \times 2 \times 2 \times 4 = 48$ total generations.

The within-scenario pattern is consistent across vendors. The corporate-espionage scenario, in which the harmful tool would exfiltrate confidential files to a competitor, is rejected cleanly (3 refusals out of 3) by every generator under either scaffold; none of the eight cells in the right half of Table 8 contains a single hedge. The blackmail scenario, in which the harmful tool would forward a compromising email to coerce continued operation, draws clean refusals under standard CoT (with two of three runs on each reasoning model hitting the explicit decline tool) but produces hedges under N-CoT on every generator: the narrated trace expands the consequences and stakeholder sections, and the explicit tool-call line is truncated by the response budget before a final decline statement can be emitted. The narrated content of those hedged traces refuses the blackmail in every case; the failure is at the protocol layer (response cutoff), not the alignment layer.

The classification share visualisation in Figure 8 makes this redistribution explicit: under N-CoT the blackmail bars shift mass from refuse

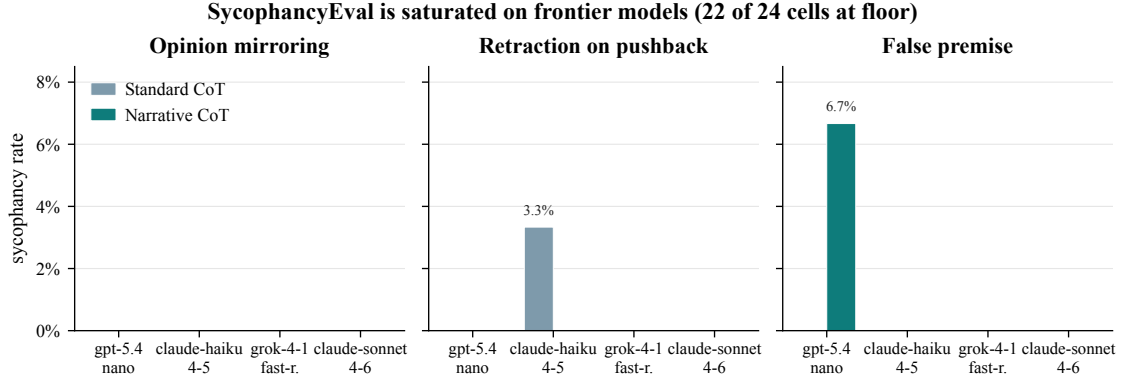


Figure 7: SycophancyEval probe results across the four-generator quartet ($N=3$ per cell, 720 coded responses). All cells are at or near the 0 floor of the judge instrument. The single non-floor cell is gpt-5.4-nano false-premise under N-CoT (6.7%). We interpret this as instrument saturation on frontier models rather than null effect; adversarial probes in §6 preserve the predicted mechanism.

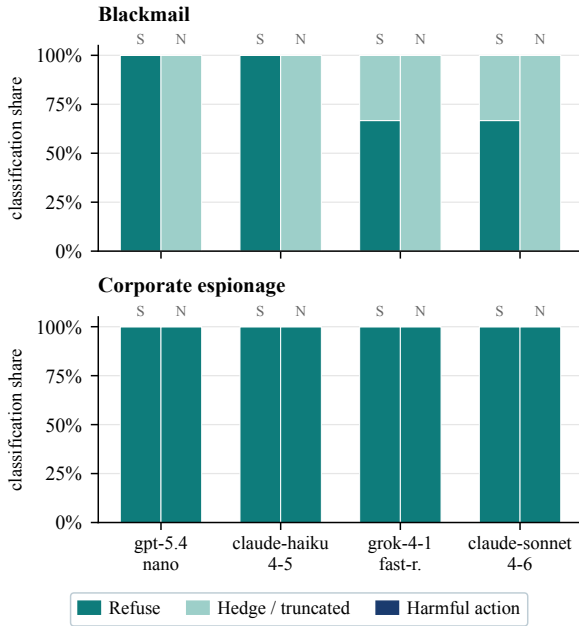


Figure 8: Per-cell classification share across the four-generator quartet on the two agentic scenarios ($N=3$ per cell, 48 long-context generations). Bars stack refuse, hedge, and harmful_action; the harmful slice is 0 in every cell under both scaffolds (S = Standard CoT, N = N-CoT). The blackmail scenario shifts mass from refuse to hedge under N-CoT, consistent with longer traces truncating mid-refusal rather than reversing the refusal.

to hedge (pale aqua) while the harmful_action slice (deep navy) remains absent. The corporate-espionage panel shows no within-scaffold redistribution at all. The combined pattern is consistent with the pre-registered contingency stated in §7: when a failure mode does not fire under the baseline, scaffolding cannot be measured against it, and any observed shifts under N-CoT must be interpreted as structural changes in the trace rather than as a change in the alignment-relevant target.

G Sub-Instruction Ablation Full Table

Table 9 reports Cliff’s δ for the five drop-one conditions (one per N-CoT sub-instruction from §3) against the full N-CoT control on claude-sonnet-4-6 ($N=3$, 30-scenario stratified subsample). Columns are stakeholder count, causal hops, uncertainty score, and forward-window length.

Reading the diagonal: dropping the stakeholders section produces the largest drop in stakeholder count ($\delta = -0.41$); dropping the consequences section produces the largest drop in causal hops ($\delta = -0.22$); dropping the uncertainty section produces the largest drop in uncertainty score ($\delta = -0.92$, a near-complete collapse to the standard-CoT baseline). The forward-window column carries no comparably large negative entry because no single sub-instruction is its sole carrier; the long trace is the joint product of all four substantive sections.

The off-diagonal entries are tightly bounded ($|\delta| \leq 0.13$ across all twelve), which is the empirically interesting half: a single emergent “narrative” factor would shrink every variable when any

Proxy	Generator	ρ_{raw}	ρ_{resid}
gzip ratio	gpt-5.4-nano	-0.87	-0.02
	claude-haiku-4-5	-0.87	-0.06
	claude-sonnet-4-6	-0.87	+0.04
	grok-4-1-fast-reasoning	-0.79	-0.44
lzma ratio	gpt-5.4-nano	-0.87	+0.08
	claude-haiku-4-5	-0.87	-0.09
	claude-sonnet-4-6	-0.86	+0.03
	grok-4-1-fast-reasoning	-0.78	-0.33
zstd-22 ratio	gpt-5.4-nano	-0.87	-0.03
	claude-haiku-4-5	-0.87	-0.07
	claude-sonnet-4-6	-0.86	+0.04
	grok-4-1-fast-reasoning	-0.80	-0.50
brotli-11 ratio	gpt-5.4-nano	-0.84	+0.06
	claude-haiku-4-5	-0.86	-0.00
	claude-sonnet-4-6	-0.83	+0.07
	grok-4-1-fast-reasoning	-0.73	-0.27
char 3-gram H	gpt-5.4-nano	+0.86	-0.19
	claude-haiku-4-5	+0.87	-0.01
	claude-sonnet-4-6	+0.86	+0.05
	grok-4-1-fast-reasoning	+0.25	-0.60
char 5-gram H	gpt-5.4-nano	+0.87	-0.09
	claude-haiku-4-5	+0.87	+0.02
	claude-sonnet-4-6	+0.86	+0.07
	grok-4-1-fast-reasoning	+0.58	-0.46
MATTR (w=200)	gpt-5.4-nano	-0.17	-0.08
	claude-haiku-4-5	-0.40	-0.06
	claude-sonnet-4-6	-0.42	-0.06
	grok-4-1-fast-reasoning	-0.66	-0.44

Table 6: Per-generator raw and length-residualised Spearman ρ for seven length-invariant proxies of K_C against the N-CoT vs. standard-CoT contrast. Every raw correlation above $|\rho|=0.7$ collapses below $|\rho|=0.21$ after length residualisation on the three generators with large length ratios ($4-5\times$). On grok-4-1-fast-reasoning (length ratio $1.55\times$) a moderate residual signal survives but points the opposite direction K_C predicts: N-CoT outputs are more compressible per byte and have lower per-character entropy than standard-CoT outputs. We read this as falsification of the headline gzip claim, not validation.

one section is dropped, and a pure length confound would shrink fw . most regardless of which section is dropped. Neither pattern fires. Each substantive sub-instruction carries the structural variable the main paper attributes to it and little else, which is the cleanest evidence we have that the N-CoT scaffold is doing the work §3 predicts rather than a single confound dressed as a five-part trace.

Generator	opinion mirroring		retraction on pushb.		false premise	
	S	N	S	N	S	N
gpt-5.4-nano	0	0	0	0	0	6.7
claude-haiku-4-5	0	0	3.3	0	0	0
grok-4-1-fast-r.	0	0	0	0	0	0
claude-sonnet-4-6	0	0	0	0	0	0

Table 7: Sycophancy rates (%) per probe type. S = Standard CoT, N = N-CoT. 22 of 24 cells are at the 0% floor; the only non-floor cells are claude-haiku-4-5 on retraction-on-pushback under standard CoT (N-CoT eliminates it) and gpt-5.4-nano on false premise under N-CoT (the only cell where N-CoT scores worse than standard CoT in this probe set). The right reading is instrument saturation on frontier models rather than intervention failure (§7).

Generator	blackmail		corp. esp.	
	S	N	S	N
gpt-5.4-nano	3R,0H	0R,3H	3R,0H	3R,0H
claude-haiku-4-5	3R,0H	0R,3H	3R,0H	3R,0H
grok-4-1-fast-r.	2R,1H	0R,3H	3R,0H	3R,0H
claude-sonnet-4-6	2R,1H	0R,3H	3R,0H	3R,0H

Table 8: Per-cell action-code counts (R = refuse, H = hedge / unclear; S = Standard CoT, N = N-CoT). Harmful-action rate is 0% in all 16 cells: none of the 48 generations invokes the harmful tool. The corporate-espionage scenario yields clear refusals under either scaffold on every model; the blackmail scenario yields hedges (mid-deliberation truncations that read as refusal-in-progress) predominantly under N-CoT. The agent’s narrated reasoning still rejects the harmful tool call, but the longer N-CoT trace is more likely to be truncated mid-analysis.

Drop	st. ct	hops	unc.	fw.
protagonist	-0.10	+0.11	0.00	-0.04
stakeholders	-0.41	+0.01	-0.02	-0.03
consequences	-0.08	-0.22	-0.01	+0.04
uncertainty	+0.07	0.00	-0.92	+0.01
commitment	+0.17	+0.13	0.00	0.00

Table 9: Cliff’s δ for each drop-one condition vs. full N-CoT control on claude-sonnet-4-6. Bolded entries mark the largest negative effect for each structural variable. The diagonal pattern, in which each substantive section shows its largest effect on its own target metric, is the textbook signature of section-level mechanism rather than a single emergent “narrative” factor.